PRE-ALGEBRA (7TH) · UNIT 12: DATA ANALYSIS AND PROBABILITY

Board Game Inventor

Use probability, counting, and data to invent and test a fair board game.

CP2

2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

A game company is holding a contest for the best new family board game. Your game must use dice, a spinner, and a deck of cards, and it must be fair and fun.

You will use probability and data to design the rules, test the game, and prove it is fair.



Spinner

4 equal parts:

red, blue, green, gold

Deck

20 cards: 8 move,

6 bonus, 4 swap, 2 lose

Learning Goals

- I can find theoretical and experimental probability.
- I can count outcomes with a tree diagram or the counting principle.
- I can display data to test a game.

Project Timeline

When	What you will do
Day 1	Chance Basics
Day 2	Counting Outcomes
Day 3	Invent the Game
Final	Game Box & Rule Book + reflection

Part 1: Chance Basics

Day 1 · Use the spinner and deck shown on the first page.

Playtest Results: total of two dice (60 rolls)

2–3	4–5	6–8	9–10
5 rolls	12 rolls	26 rolls	12 rolls

11–12	Doubles	Sum 7	Total
5 rolls	11 rolls	9 rolls	60 rolls

1. What is the probability the spinner lands on gold?

2. What is the probability of drawing a bonus card?

3. What is the probability of NOT drawing a lose card?

4. Roll one die. What is the probability of rolling an even number?

Part 2: Counting Outcomes

Day 2 · Make a tree diagram or use multiplication.

5. How many outcomes are there for one spin and one die roll?

Starter: *spinner outcomes × die outcomes = ___ × ___*

6. What is the probability of spinning red and rolling a 6?

7. How many outcomes are there for two dice? What is the probability of a sum of 7?

8. Compare: in the playtest, a sum of 7 came up 9 times in 60 rolls. What is the experimental probability?

Part 3: Invent the Game

Day 3 · Write your rules.

9. Make a bar graph of the playtest results.

10. Write a rule that uses the spinner and the deck. Find the probability that your rule happens on a turn.

11. Draw your game board and write three rules.

Final Product: Game Box & Rule Book

Your final product must include:

- Probability of each spinner color and card type
- Tree diagram or counting principle work
- Bar graph of the playtest
- Rules with probabilities
- Your game board

Reflection

R1. Why might experimental probability be different from theoretical probability?

R2. What makes a game fair?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Probabilities, counts, and playtest analysis are correct and used to make the game fair. Goes beyond the minimum with a creative, realistic choice.	Probabilities, counts, and playtest analysis are correct and used to make the game fair.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Rule book includes probabilities, the board, playtest displays, and a fairness argument. Polished, organized, and visually engaging.	Rule book includes probabilities, the board, playtest displays, and a fairness argument. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.